

ABSTRACT

The **Eye Tracking - Based Password Authentication for Enhanced Security** system is an advanced biometric security solution developed to overcome the vulnerabilities of traditional password input mechanisms, particularly threats such as shoulder-surfing, keylogging, spoofing, and unauthorized observation. The system introduces a hands-free authentication approach by leveraging real-time eye-movement patterns such as gaze direction and intentional blink sequences—to replace conventional alphanumeric password entry. Built using modern computer vision and machine learning frameworks like OpenCV and deep-learning-based eye-state classifiers, the system provides a secure and intuitive authentication workflow that ensures only the legitimate user can successfully access the protected application or service. At its core, the system captures live video through a standard webcam, detects facial landmarks, tracks the user's eye region, and identifies blink-based password inputs on a virtual interface. These captured patterns are then matched against securely stored templates using encrypted verification mechanisms, thereby preventing impersonation and enhancing overall system integrity. Additional features include support for virtual gaze-based keypads, adaptive blink detection, noise-resilient eye-state classification, and secure logging of authentication attempts. The platform emphasizes accessibility, hygiene, and usability while maintaining strict security and data-privacy considerations, making it suitable for applications in banking, enterprise systems, healthcare, and IoT environments. Although challenges such as handling variable lighting, ensuring blink-detection accuracy, and optimizing real-time processing were encountered, they were addressed through modular design practices and algorithmic refinements. This abstract highlights the system's objective, methodology, and architectural strengths, positioning Eye Tracking-Based Password Authentication as a forward-looking, tamper-resistant alternative to traditional password systems and a significant innovation in modern biometric security.